

Contexts and pseudocontexts

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In the projection lattice of a finite-dimensional Hilbert space, an orthogonal basis—a *context*—is characterised by the property that its atomic probabilities sum to one on every state. We study a weakening of this condition in which two tuples of atoms induce the same probability sum on every state. Such pairs are called *pseudocontexts*. In finite dimension this is equivalent to two distinct rank-one decompositions of the same positive operator, placing the notion naturally among finite frame-theoretic constructions. Size $k = 1$ is trivial; size $k = 2$ already admits genuinely nonorthogonal examples over $\mathbb{C}$; and, in fact, examples exist for every $k \geq 2$. For $k \geq 3$ genuine pseudocontexts also exist over $\mathbb{R}$. We present the operator formulation, give a general explicit construction, analyse the spectral structure of the associated structure operator, illustrate the symmetric $k = 3$ case with a figure, and list the remaining classification problems.

Keywords: projection lattice, pseudocontext, rank-one decomposition, structure operator, quantum contextuality

I. INTRODUCTION

Let L be the lattice of projections in a Hilbert space of finite dimension $n \geq 3$. Its atoms (minimal non-zero elements) correspond to one-dimensional subspaces, represented by nonzero vectors. States are described by probability measures $P: L \rightarrow [0, 1]$. We denote by $S(L)$ the set of all states on L .

Vectors $a_1, \dots, a_n$ form a *context* if they compose an orthogonal basis. In this case (and only in this case) they satisfy

$$\forall P \in S(L) : \sum_{i=1}^n P(a_i) = 1. \quad (1)$$

The present note concerns weaker additive identifications obtained by comparing two tuples of atoms that are not required to be orthogonal. Such *pseudocontexts* are quantum-logical shadows of orthomodular-lattice constructions of Mayet–Navara–Rogalewicz [1] and have direct physical meaning: they encode state-independent empirical equivalences between families of generally noncommuting observables. The answer is simple: pseudocontexts already exist for $k = 2$ over $\mathbb{C}$, and, more generally, for every $k \geq 2$; for $k \geq 3$ they exist over $\mathbb{R}$ as well.

II. PRELIMINARIES

Throughout, H is a complex Hilbert space with $\dim H = n \geq 3$, and $L = L(H)$ is the orthomodular lattice of its closed subspaces. Its atoms correspond to one-dimensional subspaces, or equivalently to *rays* spanned by nonzero vectors. We identify an atom with any representative unit vector a and

write $P_a = |a\rangle\langle a|$ for the orthogonal projection onto the ray it spans.

A *state* is a map $P: L \rightarrow [0, 1]$ with $P(I) = 1$ that is finitely additive on orthogonal families. By Gleason’s theorem [2], every state arises from a density operator ρ via

$$P(a) = \text{tr}(\rho P_a), \quad \rho \geq 0, \quad \text{tr} \rho = 1. \quad (2)$$

The key consequence we use repeatedly is that two self-adjoint operators that agree on every state are equal as operators.

III. CONTEXTS AND PSEUDOCONTEXTS

The operator form of (1) is elementary and is the starting point for everything that follows.

Proposition 1 (Contexts). *Unit vectors $a_1, \dots, a_n \in H$ form an orthonormal basis if and only if*

$$\sum_{i=1}^n P_{a_i} = I. \quad (3)$$

Proof. The forward implication is immediate. For the converse, fix j and evaluate (3) on the vector a_j :

$$1 = \langle a_j, I a_j \rangle = \sum_{i=1}^n |\langle a_i, a_j \rangle|^2.$$

The term with $i = j$ already contributes 1, so all others must vanish. Hence $\langle a_i, a_j \rangle = 0$ for $i \neq j$. As this holds for every j , the family is orthonormal, and being of size n in an n -dimensional space it is an orthonormal basis. $\square$

Combined with (2), Proposition 1 recovers (1) as a state-theoretic characterisation of orthogonal bases.

Definition 2 (Pseudocontext). *Let $k \geq 1$. A pseudocontext of size k is a pair of k -tuples of atoms $(a_1, \dots, a_k), (b_1, \dots, b_k)$ such that*

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- (i) each tuple contains pairwise distinct atoms,
- (ii) the two tuples are disjoint as sets of rays, so that $|\{a_1, \dots, a_k, b_1, \dots, b_k\}| = 2k$, and
- (iii) $\forall P \in S(L) : \sum_{i=1}^k P(a_i) = \sum_{i=1}^k P(b_i)$.

A pseudocontext is called *genuine* if neither k -tuple is a context.

Remark 3. Condition (i) excludes repeated atoms within a single tuple; condition (ii) ensures the two tuples are truly distinct as collections of rays. Both conditions are part of the definition; without them the notion degenerates trivially.

By (2), condition (iii) of Definition 2 is equivalent to the operator identity

$$T := \sum_{i=1}^k P_{a_i} = \sum_{i=1}^k P_{b_i}, \quad (4)$$

which we call the *structure operator* of the pseudocontext. Thus pseudocontexts are precisely pairs of distinct rank-one decompositions of a single positive operator. This places them naturally among finite frame-theoretic constructions: contexts are the special case in which the structure operator is the identity on the relevant subspace; in general, pseudocontexts are rank-one decompositions of a common positive operator. If the common operator is proportional to the identity on its support, the family is a tight frame; after normalisation on that support one obtains a POVM.

Remark 4 (Connection to SIC-POVMs). When the common structure operator satisfies $T = \frac{k}{n}I$ (tight frame on all of H), each family $\{P_{a_i}\}$ and $\{P_{b_i}\}$ scaled by n/k forms a symmetric informationally complete POVM (SIC-POVM) if $k = n^2$ and the frame vectors are equiangular [3]. Pseudocontexts therefore generalise the SIC-POVM notion by dropping both the tightness and equiangularity requirements.

IV. THE SMALLEST CASES

Proposition 5 (Size $k = 1$). *There is no pseudocontext of size $k = 1$. This holds for every $n \geq 1$.*

Proof. The condition $P(a_1) = P(b_1)$ for all $P \in S(L)$ is equivalent, via (2), to $P_{a_1} = P_{b_1}$. This rank-one identity forces a_1 and b_1 to span the same ray, contradicting the distinctness condition (ii) of Definition 2. $\square$

Remark 6 (Minimal ambient dimension). Every pseudocontext of size $k \geq 2$ can be realised inside a two-dimensional subspace, and hence embedded into $\mathbb{C}^n$ for any $n \geq 2$: all constructions in Sections IV A and V live in $\text{span}\{e_1, e_2\}$. For $k \geq 3$ the construction also works over $\mathbb{R}^n$. No pseudocontext of any size exists in dimension $n = 1$, since there every unit vector spans the unique ray, making the distinctness condition $|\{a_i\} \cup \{b_i\}| = 2k$ impossible.

A. Pseudocontexts of size $k = 2$

The $k = 2$ case requires complex scalars; we show below that no genuine real pseudocontext of size $k = 2$ exists, while the complex case allows such configurations.

Corollary 7 (Size $k = 2$, complex). *There are genuinely nonorthogonal pseudocontexts of size $k = 2$ in $\mathbb{C}^n$ for every $n \geq 2$.*

Proof. Embed $\mathbb{C}^2$ into $\mathbb{C}^n$ as the span of the first two basis vectors and set

$$u_{\pm} = \left(\frac{\sqrt{3}}{2}, \pm \frac{1}{2}, 0, \dots, 0 \right), \quad v_{\pm} = \left(\frac{\sqrt{3}}{2}, \pm \frac{i}{2}, 0, \dots, 0 \right).$$

A direct computation gives

$$P_{u_+} + P_{u_-} = P_{v_+} + P_{v_-} = \begin{pmatrix} 3/2 & 0 \\ 0 & 1/2 \end{pmatrix} \oplus 0_{n-2}.$$

The four rays are distinct: u_+ and u_- differ by a sign in the real second coordinate; v_+ and v_- differ by a sign in the imaginary second coordinate; and $u_{\pm}$ versus $v_{\pm}$ differ by a factor of i in the second coordinate, which is not a real scalar. Within each pair, $\langle u_+, u_- \rangle = -\frac{1}{4} \neq 0$ and $\langle v_+, v_- \rangle = \frac{1}{4} \neq 0$, so neither pair is orthogonal, and hence neither pair is a context. The pseudocontext is therefore genuine. $\square$

Remark 8 (Real vs. complex for $k = 2$). Genuinely nonorthogonal pseudocontexts of size $k = 2$ strictly require complex Hilbert spaces. To see why, suppose $T = P_{u_+} + P_{u_-} = P_{v_+} + P_{v_-}$ with all four vectors in $\mathbb{R}^2$. Write $u_+ = (\cos \alpha, \sin \alpha)$ and $u_- = (\cos \beta, \sin \beta)$. Then

$$T = \begin{pmatrix} \cos^2 \alpha + \cos^2 \beta & \cos \alpha \sin \alpha + \cos \beta \sin \beta \\ \cos \alpha \sin \alpha + \cos \beta \sin \beta & \sin^2 \alpha + \sin^2 \beta \end{pmatrix}.$$

For the off-diagonal entry to vanish (which is required if T is to be diagonal in the standard basis, as in the complex example above), we need $\sin(\alpha + \beta)\cos(\alpha - \beta) = 0$ modulo the constraint that u_+ and u_- are not collinear. The solutions over $\mathbb{R}$ that make T diagonal force $\{u_+, u_-\}$ to be symmetric about one of the coordinate axes, which uniquely determines the unordered pair of rays $\{u_+, u_-\}$ from T . Hence no second distinct real pair $\{v_+, v_-\}$ with the same T can exist. When T is a scalar multiple of the identity, any two-element unit-vector decomposition satisfies $\text{tr}(T) = 2$ and $T = I$, forcing the two vectors to be orthogonal (a context), so no genuine pseudocontext arises. Thus no genuine real pseudocontext of size $k = 2$ exists. This observation was made in [4], without noticing that the situation is different in complex spaces. The availability of a continuous complex phase is what makes Corollary 7 possible. For $k \geq 3$, however, genuine pseudocontexts exist over $\mathbb{R}$, as shown in Example 11.

V. PSEUDOCONTEXTS OF EVERY SIZE $k \geq 2$

Theorem 9 (Complex-phase construction). *Let H contain an orthonormal pair e_1, e_2 . Fix an integer $k \geq 2$, let $\omega = e^{2\pi i/k}$,*

choose $\alpha \in (0, \pi/2)$, and choose $\phi \in \mathbb{R}$ such that $e^{i\phi}$ is not a k th root of unity. Define unit vectors

$$\begin{aligned} a_j &= \cos \alpha e_1 + \sin \alpha \omega^j e_2, \\ b_j &= \cos \alpha e_1 + \sin \alpha e^{i\phi} \omega^j e_2, \quad j = 0, \dots, k-1. \end{aligned}$$

Then the $2k$ rays spanned by the a_j and b_j are pairwise distinct, and

$$\sum_{j=0}^{k-1} P_{a_j} = \sum_{j=0}^{k-1} P_{b_j} = k(\cos^2 \alpha P_{e_1} + \sin^2 \alpha P_{e_2}).$$

Hence $(a_0, \dots, a_{k-1})$ and $(b_0, \dots, b_{k-1})$ form a pseudocontext of size k . If $\alpha \neq \pi/4$, then the pseudocontext is genuine.

Proof. Write $c = \cos \alpha$ and $s = \sin \alpha$. For each j ,

$$\begin{aligned} P_{a_j} &= c^2 P_{e_1} + s^2 P_{e_2} + cs(\omega^{-j}|e_1\rangle\langle e_2| + \omega^j|e_2\rangle\langle e_1|), \\ P_{b_j} &= c^2 P_{e_1} + s^2 P_{e_2} + cs(e^{-i\phi}\omega^{-j}|e_1\rangle\langle e_2| + e^{i\phi}\omega^j|e_2\rangle\langle e_1|). \end{aligned}$$

Summing over $j = 0, \dots, k-1$ and using $\sum_{j=0}^{k-1} \omega^j = 0$ cancels all off-diagonal terms, giving

$$\sum_{j=0}^{k-1} P_{a_j} = \sum_{j=0}^{k-1} P_{b_j} = k(c^2 P_{e_1} + s^2 P_{e_2}).$$

Distinctness. If $a_j = \lambda a_\ell$ for some scalar λ , then comparing first coordinates (which equal $c \neq 0$) gives $\lambda = 1$, and then $\omega^j = \omega^\ell$, so $j = \ell$. Hence the a_j are pairwise distinct rays; the identical argument applies to the b_j . If $a_j = \lambda b_\ell$, then $\lambda = 1$ and $\omega^j = e^{i\phi} \omega^\ell$, i.e. $e^{i\phi} = \omega^{j-\ell}$, contradicting the assumption that $e^{i\phi}$ is not a k th root of unity. Hence no a_j coincides with any b_ℓ .

Genuineness. For $j \neq \ell$ within the a -tuple,

$$\langle a_j, a_\ell \rangle = c^2 + s^2 \omega^{\ell-j}.$$

This vanishes only if $c^2 = s^2$ (i.e. $\alpha = \pi/4$) and $\omega^{\ell-j} = -1$. So for $\alpha \neq \pi/4$ no two distinct vectors in the a -tuple are orthogonal. An identical computation gives $\langle b_j, b_\ell \rangle = c^2 + s^2 \omega^{\ell-j}$, so the same conclusion holds for the b -tuple. Hence neither tuple is a context when $\alpha \neq \pi/4$. $\square$

Remark 10. At the special value $\alpha = \pi/4$, the structure operator becomes $\frac{k}{2} I_{\text{span}\{e_1, e_2\}}$, so the construction yields a unit-norm tight frame on the two-dimensional support. For $k = 2$ this reduces to a context; for $k \geq 3$ it gives a highly symmetric pseudocontext (cf. Remark 12).

VI. A SYMMETRIC $k = 3$ EXAMPLE

The case $k = 3$ admits a very transparent planar representation; it is the finite-dimensional analogue of the Mercedes-Benz frame.

Example 11 (Mercedes-Benz type pseudocontext). Fix a two-dimensional subspace $V \subset H$ with orthonormal basis (e_1, e_2) , and fix an angle θ satisfying $\theta \neq 0$, $\theta \neq \pi/3$, and $\theta \not\equiv 2\pi/3$ modulo π . Put, for $j = 0, 1, 2$,

$$a_j = \cos\left(\frac{2\pi j}{3}\right) e_1 + \sin\left(\frac{2\pi j}{3}\right) e_2,$$

$$b_j = \cos\left(\frac{2\pi j}{3} + \theta\right) e_1 + \sin\left(\frac{2\pi j}{3} + \theta\right) e_2.$$

Then the six rays are pairwise distinct, and

$$\sum_{j=0}^2 P_{a_j} = \sum_{j=0}^2 P_{b_j} = \frac{3}{2} I_V.$$

Thus $\{a_0, a_1, a_2\}$ and $\{b_0, b_1, b_2\}$ form a genuine pseudocontext of size $k = 3$ over $\mathbb{R}$.

Proof. In the basis (e_1, e_2) , each projector has the form

$$P_{\cos \varphi e_1 + \sin \varphi e_2} = \begin{pmatrix} \cos^2 \varphi & \cos \varphi \sin \varphi \\ \cos \varphi \sin \varphi & \sin^2 \varphi \end{pmatrix}.$$

For $\varphi_j = 2\pi j/3$ and for $\varphi_j = \theta + 2\pi j/3$, the trigonometric identities

$$\sum_{j=0}^2 \cos(2\varphi_j) = 0, \quad \sum_{j=0}^2 \sin(2\varphi_j) = 0,$$

which hold for any arithmetic progression of three angles spaced $2\pi/3$ apart, imply that the off-diagonal terms cancel while the diagonal terms each sum to $3/2$. Hence both sums equal $\frac{3}{2} I_V$.

The three excluded residue classes modulo π account for all collisions between the two triples:

- $\theta \equiv 0 \pmod{\pi}$: every b_j coincides with a_j or $-a_j$, giving the same ray.
- $\theta \equiv \pi/3 \pmod{\pi}$: the b -triple is a cyclic permutation of the a -triple as a set of rays.
- $\theta \equiv 2\pi/3 \pmod{\pi}$: the b -triple is again a permutation of the a -triple.

For all other values of θ the six rays are pairwise distinct. Genuineness holds because for generic θ no two vectors within the same triple are orthogonal (the angle between consecutive vectors in each triple is $2\pi/3 \neq \pi/2$), so neither triple is a context. $\square$

Remark 12 (Complex trine description). Example 11 has a compact complex analogue. Let $\omega = e^{2\pi i/3}$ and embed V into $\mathbb{C}^n$ as $\text{span}\{e_1, e_2\}$. The *trines*

$$u_j = \frac{1}{\sqrt{2}}(1, \omega^j, 0, \dots, 0), \quad v_j = \frac{1}{\sqrt{2}}(1, e^{i\phi} \omega^j, 0, \dots, 0), \quad j = 0, 1, 2,$$

yield the same phenomenon after a change of basis in V . The identity $1 + \omega + \omega^2 = 0$ makes the off-diagonal entries of $\sum_j P_{u_j}$ cancel, giving the common structure operator $\frac{3}{2} I_V$. This form is most natural in frame-theoretic discussions and in connection with the Weyl-Heisenberg group.

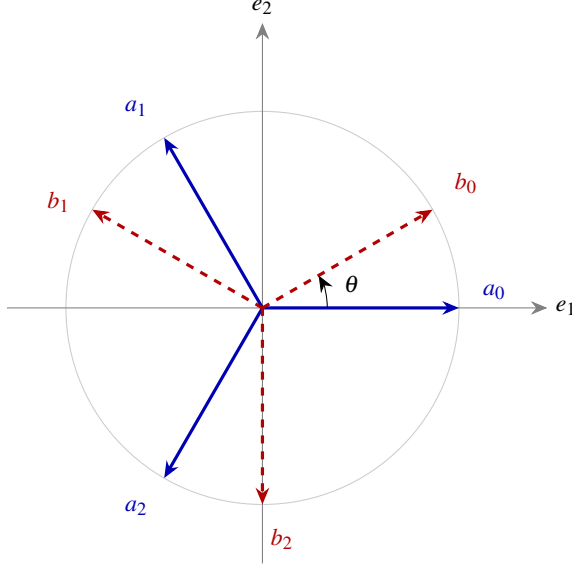


FIG. 1. The Mercedes–Benz pseudocontext of Example 11, drawn for $\theta = \pi/6$. The solid blue rays a_0, a_1, a_2 and the dashed red rays b_0, b_1, b_2 are two distinct unordered triples in the plane V , yet the associated rank-one projectors have the same sum $\frac{3}{2}I_V$.

Remark 13 (Nondegenerate structure operators). The trine family is highly symmetric: its structure operator is a scalar multiple of the identity on its support. More complicated examples with nondegenerate structure operators occur already in dimension three; see, for instance, the coordinatisations discussed in [5, 6]. The main difference is that the pseudocontext in Example 11 has only one constant value in the probability sum, while the advanced constructions admit a larger range of values, still respecting the equality. Such examples are useful when one wants to move beyond purely symmetric frame orbits.

Similar constructions [1] were known when L was a more general orthomodular lattice [7]. Some of these lattices can be represented as subsets of a Hilbert lattice [5, 6].

VII. SPECTRAL STRUCTURE AND CANONICAL FORMS

We analyse the general properties of the structure operator $T = \sum_{i=1}^k P_{a_i}$ associated with a k -tuple of unit vectors $(a_1, \dots, a_k)$ in H .

A. Gram representation and spectral equivalence

Definition 14 (Analysis operator and Gram matrix). Let $A: \mathbb{C}^k \rightarrow H$ be defined by $Ae_i = a_i$, where (e_i) is the standard basis of $\mathbb{C}^k$. The analysis operator is $A^\dagger: H \rightarrow \mathbb{C}^k$, and the Gram matrix is

$$G := A^\dagger A = (\langle a_i, a_j \rangle)_{i,j=1}^k.$$

Proposition 15 (Frame operator factorisation). *The structure operator admits the factorisation $T = AA^\dagger$, and $G = A^\dagger A$.*

Proposition 16 (Spectral equivalence). *The nonzero spectra of T and G coincide (including multiplicities):*

$$\sigma(T) \setminus \{0\} = \sigma(G) \setminus \{0\}.$$

In particular, $\text{rank}(T) = \text{rank}(G) \leq \min(k, n)$.

Proof. This is the standard singular-value correspondence between $AA^\dagger$ and $A^\dagger A$: if $AA^\dagger v = \lambda v$ with $\lambda \neq 0$, then $A^\dagger A(A^\dagger v) = \lambda(A^\dagger v)$ and $A^\dagger v \neq 0$. $\square$

B. Spectral constraints

Proposition 17 (Trace identities). *The structure operator satisfies*

$$\text{tr}(T) = k, \quad \text{tr}(T^2) = \sum_{i,j=1}^k |\langle a_i, a_j \rangle|^2.$$

The diagonal terms $i = j$ each contribute 1, giving $\text{tr}(T^2) \geq k$, with equality if and only if the vectors are mutually orthogonal.

Proof. $\text{tr}(T) = \sum_i \text{tr}(P_{a_i}) = k$. For the second identity, $\text{tr}(T^2) = \sum_{i,j} \text{tr}(P_{a_i} P_{a_j}) = \sum_{i,j} |\langle a_i, a_j \rangle|^2$. Diagonal terms contribute k ; off-diagonal terms are non-negative, and vanish simultaneously if and only if all off-diagonal inner products are zero. $\square$

Proposition 18 (Non-orthogonality and spectral spread). *If the vectors (a_i) are not mutually orthogonal, then $\text{tr}(T^2) > k$, the spectrum of T is non-flat, and $\lambda_{\max}(T) > 1$.*

Proposition 19 (Majorisation constraint). *Suppose $k \leq n$. Let $\lambda(T) = (\lambda_1, \dots, \lambda_n)$ be the eigenvalues of T in decreasing order (padded with zeros to length n), and let $d = (1, \dots, 1, 0, \dots, 0) \in \mathbb{R}^n$ have k ones and $n - k$ zeros. Then $\lambda(T) \succ d$.*

Proof. The diagonal entries of G are $G_{ii} = \|a_i\|^2 = 1$, so the diagonal of G is $(1, \dots, 1) \in \mathbb{R}^k$. By the Schur–Horn theorem, $\lambda(G) \succ (1, \dots, 1)$. Since G and T share the same nonzero eigenvalues (Proposition 16), padding $\lambda(G)$ with $n - k$ zeros gives $\lambda(T)$, and $\lambda(T) \succ d$ follows.

Overcomplete case $k > n$. When $k > n$ the Gram matrix G is $k \times k$ with rank at most n ; the majorisation is then best stated for the k -dimensional eigenvalue vector of G against $(1, \dots, 1) \in \mathbb{R}^k$, and the Schur–Horn argument applies to G directly. $\square$

Proposition 20 (Operator bounds). *The structure operator satisfies $0 \leq T \leq kI$. Moreover, if $\mu := \max_{i \neq j} |\langle a_i, a_j \rangle|$, then*

$$\lambda_{\max}(T) \leq 1 + (k - 1)\mu.$$

Proof. Positivity $T \geq 0$ is immediate, and $T \leq kI$ since each $P_{a_i} \leq I$. For the eigenvalue bound, apply the Geršgorin circle theorem to G : since $G_{ii} = 1$ and $|G_{ij}| \leq \mu$ for $i \neq j$, every eigenvalue of G lies in a disk centred at 1 with radius at most $(k - 1)\mu$. Since the nonzero eigenvalues of T and G coincide, $\lambda_{\max}(T) \leq 1 + (k - 1)\mu$. $\square$

C. Commutant and symmetry

Definition 21 (Commutant). The *unitary commutant* of T is $\mathcal{C}(T) := \{U \in U(H) : UT = TU\}$.

Proposition 22 (Structure of the commutant). Let $T = \sum_{\alpha} \lambda_{\alpha} P_{\alpha}$ be the spectral decomposition of T with distinct eigenvalues λ_{α} and orthogonal projections P_{α} of ranks m_{α} . Then $\mathcal{C}(T) \cong \prod_{\alpha} U(m_{\alpha})$.

Remark 23. Spectral degeneracies ($m_{\alpha} > 1$) give rise to continuous families of distinct rank-one decompositions of T (as in tight frames), whereas simple spectrum leads to strong rigidity. This explains why the tight-frame case (Remark 10) is so rich in pseudocontexts.

D. Canonical normal form

Proposition 24 (Spectral normal form). Every structure operator is unitarily equivalent to $\text{diag}(\lambda_1, \dots, \lambda_r, 0, \dots, 0)$ with $\lambda_i > 0$, $\sum_{i=1}^r \lambda_i = k$, and $r = \text{rank}(T)$.

E. Canonical parametrisation of all decompositions

Theorem 25 (Canonical decomposition). Let T be a positive operator of rank r with spectral decomposition $T = \sum_{i=1}^r \lambda_i |e_i\rangle\langle e_i|$, $\lambda_i > 0$, and set $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_r)$.

Every rank-one decomposition $T = \sum_{j=1}^k P_{a_j}$ with $\|a_j\| = 1$ corresponds to a matrix $C \in \mathbb{C}^{r \times k}$ satisfying

$$CC^{\dagger} = \Lambda, \quad \|c_j\|^2 = 1 \text{ for all } j, \quad (5)$$

where c_j is the j -th column of C , via $a_j = \sum_{i=1}^r c_{ij} e_i$. Conversely, every such C defines a valid decomposition.

Proof. Let $T = AA^{\dagger}$ with columns a_j . In the eigenbasis of T , write $A = UC$ where U is the unitary diagonalising T on its support. Then $CC^{\dagger} = U^{\dagger} T U|_{\text{supp}(T)} = \Lambda$, and $\|c_j\| = \|a_j\| = 1$. The converse is a direct computation. $\square$

Corollary 26 (Unitary freedom). Among all rank-one decompositions of T into k (not necessarily unit-norm) vectors, any two are related by $A \mapsto AU$ for some $U \in U(k)$. Imposing the unit-norm constraint (5) cuts out a submanifold of this $U(k)$ -orbit; the action of $U(k)$ is not transitive on unit-norm decompositions in general.

F. Characterisation of pseudocontexts

Proposition 27 (Operator equivalence criterion). Two k -tuples (a_i) and (b_i) of distinct rays form a pseudocontext if and only if their matrices C_A, C_B in the eigenbasis of T both satisfy (5):

$$C_A C_A^{\dagger} = C_B C_B^{\dagger} = \Lambda.$$

Remark 28. Pseudocontexts correspond to distinct points in the space $\{C \in \mathbb{C}^{r \times k} : CC^{\dagger} = \Lambda, \|c_j\| = 1\} / \sim$, where $\sim$ identifies decompositions that yield the same set of rays.

G. Extremal spectral bounds

Proposition 29 (Rayleigh bounds). For any density operator ρ ,

$$\lambda_{\min}(T) \leq \text{tr}(\rho T) \leq \lambda_{\max}(T).$$

Both bounds are tight, achieved by pure states on the corresponding eigenvectors of T .

H. Examples of nondegenerate spectra

Explicit three-dimensional implementations of nondegenerate structure operators were derived in [5].

- **15-atom hypergraph (T_1).** A pseudocontext of size $k = 3$ from a generalised firefly logic yields

$$\lambda_1 = 2, \quad \lambda_2 = \frac{7+\sqrt{21}}{14} \approx 0.827, \quad \lambda_3 = \frac{7-\sqrt{21}}{14} \approx 0.173,$$

with $\text{tr}(T_1) = 3$ and $\lambda_{\max} = 2 > 1$, consistent with Proposition 18.

- **36-atom hypergraph (T_2).** A larger true-implies-false gadget gives $T_2 = \text{diag}(1/5, 1/5, 13/5)$ with $\text{tr}(T_2) = 3$ and $\lambda_{\max} = 13/5 > 1$.

VIII. PHYSICAL INTERPRETATION AND CONSTRAINTS

A pseudocontext $(\{a_i\}, \{b_i\})$ implements a state-independent empirical equivalence between two families of rank-one observables $A_i = P_{a_i}$ and $B_i = P_{b_i}$. Although the operators within each family are generally noncommuting, the two linear functionals

$$\rho \mapsto \sum_i \text{tr}(\rho A_i), \quad \rho \mapsto \sum_i \text{tr}(\rho B_i)$$

agree on the whole state space.

The spectral decomposition of T establishes absolute quantum limits on these probability sums. By Proposition 29,

$$\lambda_{\min}(T) \leq \sum_{i=1}^k P(a_i) \leq \lambda_{\max}(T). \quad (6)$$

These bounds are tight.

For the 36-atom hypergraph example [5], the structure operator $T_2 = \text{diag}(1/5, 1/5, 13/5)$ confines the quantum probability sum to $[1/5, 13/5] = [0.2, 2.6]$. A classical hidden-variable model based on two-valued $(\{0, 1\})$ truth assignments to projections—subject to the consistency requirement that exactly one atom in each orthogonal basis receives the value 1—would allow the sum to range over $\{0, 1, 2, 3\}$, which includes the extreme values 0 and 3. Because T_2 has no eigenvalue equal to 0 or 3, quantum mechanics forbids these extremes; the quantum system cannot reach them. This discrepancy is a signature of Kochen–Specker-type contextuality [8, 9].

This state-independent additive equivalence is related to, but distinct from, the operational notion of contextuality introduced by Spekkens [10], which applies to arbitrary experimental procedures (preparations, transformations, and unsharp measurements) rather than to sharp projective measurements alone. A precise comparison is beyond the scope of this note.

In the language of quantum information, $\{A_i\}$ and $\{B_i\}$ are distinct rank-one decompositions of the same positive operator. When this operator is proportional to the identity on its support, the families form tight frames; after normalisation they yield POVMs.

IX. OPEN PROBLEMS

Open Problem (Classification). Classify, up to unitary equivalence, all pseudocontexts of size k in $\mathbb{C}^n$. Even for small (k, n) , the orbit structure is not yet transparent.

Open Problem (Lattice representability). Characterise which orthomodular lattices carrying a pseudocontext-type additive identification admit a faithful embedding into some $L(\mathbb{C}^n)$, sharpening the partial results of [5, 6].

Open Problem (Frame-symmetry characterisation). Can one

characterise pseudocontexts purely in terms of symmetries of the underlying rank-one frame, for example in terms of unitaries commuting with the structure operator and permuting the atoms of the two decompositions?

Open Problem (Minimal ambient dimension). Although every pseudocontext of size $k \geq 2$ can be realised in dimension $n = 2$ over $\mathbb{C}$, what is the minimal n for which *all* pseudocontexts of a given combinatorial type occur? For Kochen–Specker configurations this question has a long history; the pseudocontext analogue is open.

Open Problem (Minimal size over $\mathbb{R}$). Example 11 shows that genuine pseudocontexts exist over $\mathbb{R}$ for $k = 3$. Remark 8 shows they do not exist for $k = 2$. It remains to give a complete proof that $k = 3$ is the minimal size for genuine real pseudocontexts, and to classify all such minimal examples up to orthogonal equivalence.

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